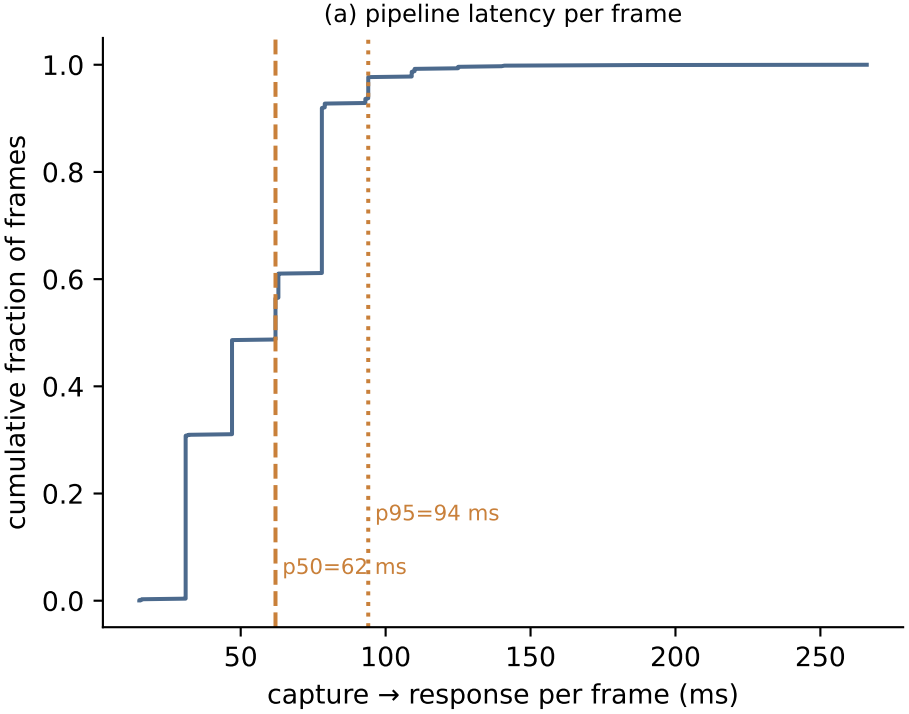
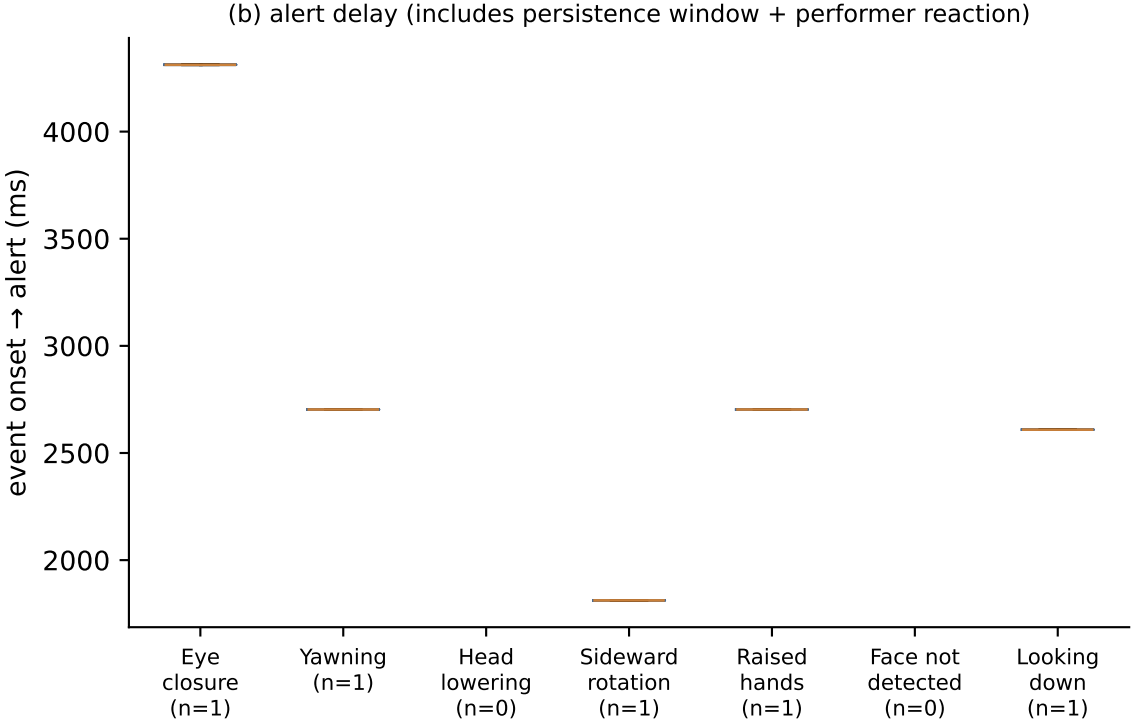

(a) pipeline latency per frame
cumulative fraction of frames
1.0
0.8
0.6
0.4
0.2
0.0
p95=94 ms
p50=62 ms
50
100
150
200
250
capture → response per frame (ms)


(b) alert delay (includes persistence window + performer reaction)
event onset → alert (ms)
4000
3500
3000
2500
2000
Eye closure (n=1)
Yawning (n=1)
Head lowering (n=0)
Sideward rotation (n=1)
Raised hands (n=1)
Face not detected (n=0)
Looking down (n=1)